How similar must the two conditional states be?

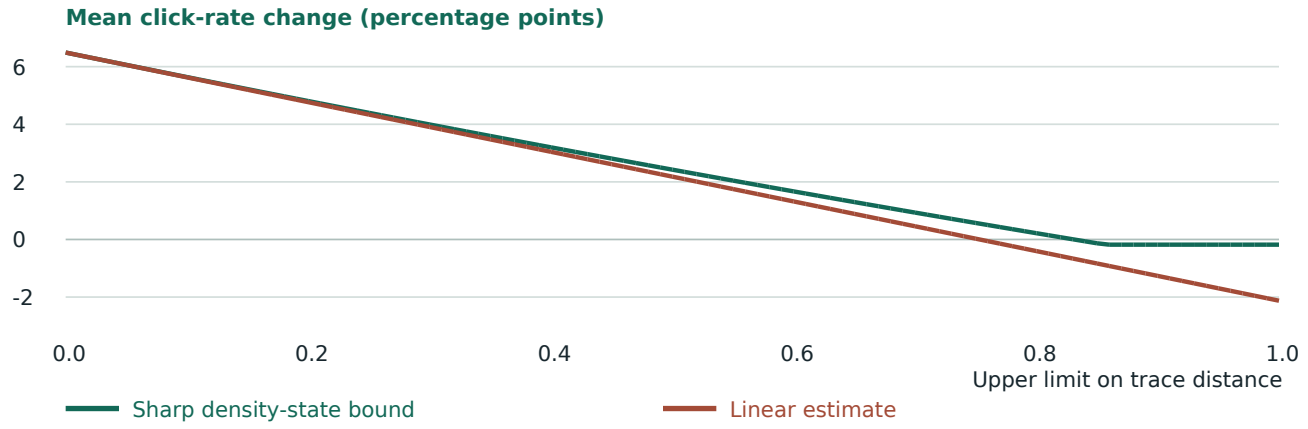
Research direction and responsibility · Kirtus Justus. Mathematical and computational assistance · ChatGPT (OpenAI). Port Coquitlam, British Columbia, Canada · 7 November 2026.

Consider equal groups at offsets +60° and −60° from a commanded joint reflection after history 111. Let $D=\tfrac{1}{2}\|\rho_{+}-\rho_{-}\|_{1}$. Given an upper limit $D\le\tau$, what change in click probability must every compatible pair produce? The source gives a valid linear estimate. The exact minimum follows by intersecting the readout positivity constraints with the distance constraint.

> For general density states, the sharp strict-rise condition based on the cap alone is τ below 0.830970598587409… . At that value zero is attainable. For two normalized trace-preserving channel states with a maximally entangled match, the minimum is still strictly positive at that same cap. Its unconstrained sign at 60° remains open.

Distance cap τ	Source linear bound (pp)	Sharp minimum (pp)
0	+6.49019	+6.49019
0.50	+2.18177	+2.42395
0.80	−0.40328	+0.21771
0.84	−0.74795	−0.06312

Rounded to nearest. All rows concern model populations and the same detector across groups. The sharp column allows any detector effect consistent with h. It also applies in the specified sixteen-dimensional probe, and the examples attaining the minima use only three dimensions.



No trace distance, gate-path offset distribution, or history-matched phase result was measured in the supplied files. These curves describe what a separately justified upper limit could establish.

The exact distance-constrained minimum

Common rank-one $P=|p\rangle\langle p|$, binary effect E, $R=I-P$ and $U(\varphi)=P+\exp(i\varphi)R$. The direct and phase arms use matched weights. The match population s and ordinary click rate q are pooled.

Set $s=131/320$, $q=81711338569043/224995905216000$, $h=\langle p|E|p\rangle=0.99$, $a=sh$, $t=a-q$, $g=\sqrt{[h(1-h)]}$. In group j let $b_j=\mathrm{Tr}(E\,R\rho_j R)$ and $c_j=\mathrm{Tr}(E\,P\rho_j R)$. Write $b=(b_+ +b_-)/2$ and $c'=(c_+ +\mathrm{conj}(c_-))/2=-(t+b)/2+iz/2$. Direct expansion gives $\Delta=(1+\cos\delta)(t+b)-\sin\delta\, z$. Take $0\le\delta\le\pi$.

Positivity of E and its complement yields $|c_j|^2\le\min\{hs_jb_j,(1-h)s_j(1-s_j-b_j)\}$. Applying triangle and Cauchy–Schwarz after conjugating the negative group gives the same two inequalities for c' with pooled s,b. In particular $(b-u)^2+z^2\le R_1^2$, with $u=a+q$ and $R_1=2\sqrt{(aq)}$. The two circles meet at $b_0=(1-s)(1-h)$, $z=\pm z_0$ where $z_0=\sqrt{[4ab_0-(t+b_0)^2]}$. The first circle's allowed height increases up to b_0; the second circle's height decreases beyond it. The maximum $|z|$ is z_0.

Put $G=REP$ and $\sigma=(\rho_+-\rho_-)/2$. Then $z=\mathrm{Tr}[i(G\dagger-G)\sigma]$, so $|z|\le\|G\|D\le g\tau$. Here $\|G\|^2=\langle p|E^2|p\rangle-h^2\le h(1-h)$. Replacing b by its absolute minimum $u-R_1$ and z by $g\tau$ reproduces the submitted linear bound. The first disk gives a stronger simultaneous constraint: at a given $0\le z\le z_0$ the smallest feasible b is $u-\sqrt{(R_1^2-z^2)}$. This lies in the second disk because $b\le b_0$.

> Choose $z=\min\{g\tau,\, z_0,\, R_1\sin(\delta/2)\}$ and $b=u-\sqrt{(R_1^2-z^2)}$. Then $\Delta_{\min}=(1+\cos\delta)(t+b)-\sin\delta\, z$. This follows by differentiating the scalar objective on $0\le z\le\min(g\tau,z_0)$; at $\delta=\pi$ the contrast is identically zero.

Attainment. In basis p,v,w use $E=|e\rangle\langle e|$ with $e=\sqrt{h}\,p+\sqrt{(1-h)}v$. Both states have diagonals $s,b/(1-h),1-s-b/(1-h)$. Only the p,v entry and its conjugate are off diagonal. Set $\rho_{pv}=[-(t+b)\pm iz]/(2g)$. The disk bound gives positivity, and $b\le b_0$ ensures nonnegative spectator mass. Direct multiplication gives the required q in each group and the displayed contrast. The state difference has eigenvalues $\pm z/g$ and zero, so $D=z/g\le\tau$.

At $\delta=60°$ the first zero occurs on the first circle with $z=\sqrt{3}(t+b)$. Thus $(t+b)^2=ab$ at the smaller root $b_\star=(a-2t-\sqrt{[a(a-4t)]})/2$. It obeys $0<b_\star<b_0$. The exact critical cap is $\tau_\star=\sqrt{3}(t+b_\star)/g$. Both the cap and the angular sign threshold refer to exact expressions; printed decimals are approximations.

Why channels cannot attain that zero

Here P is a maximally entangled unitary match and both normalized states are Choi states of trace-preserving maps on a four-dimensional system. The common E is otherwise unrestricted.

At $\tau=\tau^\star$ a nonpositive contrast would attain the unique general-state optimum $(b^\star,g\tau^\star)$. This requires $D=\tau^\star$, $\|REP\|=g$ and equality in the trace-norm inequality for z. Put $v=REPp/g$. From $(E-E^2)p=0$ one gets $Ev=gp+(1-h)v$. The Hermitian observable $i(G^\dagger-G)$ has simple extreme eigenvectors $(p\pm iv)/\sqrt{2}$. Equality places the positive and negative parts of σ on these opposite eigenvectors. Tr σ=0 fixes their weights to be equal. Thus the group difference is purely imaginary off-diagonal in span{p,v}, with zero difference on all other blocks.

The compression of either group to p,v has the same diagonals s,k. Write its upper off-diagonal entry as $x+i(m\pm n)$; then $gx=-(t+b^\star)/2$ and $gn=z/2$ (up to sign convention). Positivity gives $sk\geq x^2+m^2+n^2$ by averaging the two determinants. Also $b^\star\geq(1-h)k$. Equality on disk 1 gives $x^2+n^2=sb^\star/(1-h)$. Hence $k=b^\star/(1-h)$, $m=0$, and both compressed blocks have rank one. Any coupling to the spectator space must have columns in that block's range. The coupling is the same in the two groups, yet the two ranges are distinct, so the coupling vanishes.

In the output-first Choi convention let $p=|I_4\rangle$ and $v=|V\rangle$. Tr V=0 and $\mathrm{Tr}(V^\dagger V)=4$. The vanishing output partial trace of the imaginary off-diagonal state difference forces $V=V^\dagger$. Each group contains a positive Choi summand with operator $K=\sqrt{s}\,I+\alpha V$, where $|\alpha|^2=k$ and $2\sqrt{s}\,\mathrm{Re}\,\alpha=-(t+b^\star)/g$. Trace preservation of the full state therefore requires $K^\dagger K\leq I$. Each eigenvalue λ of V obeys $k\lambda^2-(t+b^\star)\lambda/g-(1-s)\leq 0$.

> Both roots lie inside $[-0.684,1.537]$. Four real eigenvalues in that box with zero sum have mean square at most 0.8316605. The normalized Choi vector requires mean square one. This contradiction excludes zero at the critical cap.

For the numerical box, interval arithmetic gives a positive polynomial at each endpoint and a negative value at zero. Maximizing the convex sum of squares over the box with zero sum puts at least three coordinates at endpoints. The only admissible pattern, up to permutation, is $(-0.684,-0.684,1.537,-0.169)$. The common-effect and normalized-channel feasible set is compact, so the minimum is strictly positive at $\tau^\star$. This proof supplies neither its size nor its sign without the distance cap.

Calibration and the separate purity check

No numerical distance, fidelity, second-output expectation or offset response has been assigned to an apparatus.

A possible calibration target

The norm argument uses only the signed quadrature $z=\mathrm{Tr}[i(G^\dagger-G)(\rho_+-\rho_-)/2]$. If the effect and projector can be characterized, this Hermitian observable can in principle be measured on each matched group. A certified upper limit on z then cuts the first disk directly. A trace-distance upper limit controls every observable; in particular it gives $z\le g\tau$. Conversely, agreement on a single unrelated observable generally does not upper-bound the distance between two states.

For the two groups, the source's sufficient linear threshold $D<0.753198530485524\ldots$ checks out, as does the adverse example $D\approx0.85687405$. Using squared fidelity Fsq and $D\le\sqrt{(1-\mathrm{Fsq})}$, $\mathrm{Fsq}>1-\tau_\star^2\approx0.30948786$ is sufficient with the supplied inputs [1]. This last conversion is not asserted to be a sharp fidelity threshold. Common effect, equal weights, matched ordinary arms and the same joint phase action are still required. If s,h,q or the angles are uncertain, the lower bound has to be minimized over their simultaneously justified region.

What the second-qubit Y expectation identifies

This is a different experiment on the particular two-sign unitary family. Its orthogonal Choi components have populations $1-\ell$ and ℓ and off-diagonal $t'\sqrt{[\ell(1-\ell)]}$. The sign of that off-diagonal disappears from the first-qubit output on $|00\rangle$ when the second qubit is traced out. For the calibrated auxiliary unitary N, $a_0=\langle0|N|0\rangle>0$. The second-qubit marginal gives $y=2a_0t'\sqrt{[\ell(1-\ell)]}$, so the purity along this family is $(1-\ell)^2+\ell^2+y^2/(2a_0^2)$. At $\ell=0.01$ the endpoints give $\pm0.10229823305\ldots$ and purity one; their equal mixture has $y=0$ and purity 0.9802. This measurement is local to the second output. A two-qubit correlation measurement is unnecessary for this identity.

At exact f=r the previously derived first-qubit transverse identity $d^2+r^2+o^2/2$ is a separate boundary result. Its inference needs a normalized trace-preserving Choi state and transfer to the different product preparation. The linear and sharp distance guarantees on pp. 1–2 concern general states. Only the additional exclusion on p. 3 requires the Choi marginal. White et al. give an experimental framework for examining intervention-dependent dynamics [2]. None of the submitted results measures this gate path.

Mathematical checks
The accompanying calculation encloses the contrast threshold, compares Born probabilities and trace distances with the formula, tests pooled-state controls and verifies the spectral exclusion box. These deterministic checks concern model populations.

K. Justus, Angle Moments and Variance, accompanying mathematical notes.

[1] J. Watrous, The Theory of Quantum Information, ch. 3, Theorem 3.33. Source ↗

[2] G. A. L. White et al., Non-Markovian Quantum Process Tomography (2022). Source ↗